

AI-FITTRACK

PHASE 4- Project Planning

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1. PROJECT PLANNING OVERVIEW

Project Planning defines how the verified AI FitTrack requirements and design are organized into an executable academic project workflow. The plan establishes scope, responsibilities, development sequence, testing preparation, documentation activities, demonstration preparation, and risk or issue handling.

2. PROJECT SCOPE FOR PLANNING

The planned implementation is the assigned AI FitTrack backend REST API. The planning baseline includes Node.js and Express.js, MongoDB with Mongoose, JWT authentication, bcryptjs password protection, workout CRUD and search operations, Google Gemini workout recommendations and fitness insights, API testing, documentation, GitHub organization, and final demonstration.

The plan does not schedule a separate React frontend, administrator dashboard, notification module, or additional fitness collections as current implementation work. Such items remain outside the verified project scope unless formally introduced later as future extensions.

3. DEVELOPMENT STRATEGY

Strategy Area	Planned Approach
Implementation Basis	Use the supplied AI FitTrack project implementation and its documented architecture as the development baseline.
Development Style	Proceed incrementally from project setup and configuration to backend modules, database integration, AI integration, testing, and evidence preparation.
Architecture Approach	Maintain separation between routes, controllers, middleware, models, services, configuration, and server/application files.
Database Approach	Use MongoDB/Mongoose with the verified User and Workout models and the User-to-Workout relationship.
Authentication Approach	Configure registration and login first, then verify JWT-protected operations before testing dependent modules.
AI Approach	Configure the Gemini API through environment variables and verify recommendation and fitness-insight operations after core API functions are working.
Testing Approach	Test authentication first, then workout APIs, then AI APIs, followed by error and regression checks.
Evidence Approach	Capture meaningful execution, API, database, and output evidence while completing the corresponding Skill Wallet tasks.
Repository Approach	Maintain a public GitHub repository with phase-wise documentation and source code while excluding secrets and node_modules.

4. TEAM RESPONSIBILITIES AND TASK ALLOCATION

No.	Team Member	Planned Responsibility
1	CHANDRA SEKAR G.	Team Leader – project integration, backend coordination, AI integration, final testing and debugging, GitHub and demonstration coordination.
2	CHANDRU G T	Backend Development and API Integration – route configuration, controller implementation, middleware integration, API functionality and debugging support.
3	CHANDRU M	MongoDB Configuration and Database Design – MongoDB Atlas, User/Workout schemas, relationships, validation and database verification.
4	BOOPATHI B L	API Testing and Verification – authentication, workout CRUD/search, AI endpoints, test evidence and regression testing.
5	CHANDAN KUMAR B	Architecture, Documentation and Execution Evidence – diagrams, project structure documentation, screenshots, README and repository organization.

5. SKILL WALLET WORK BREAKDOWN

Epic	Tasks Included	Planning Focus
1. Project Architecture	1. Technical Architecture; 2. ER Diagram; 3. Features; 4. Roles and Responsibilities; 5. User Flow; 6. MVC Pattern	Establish the system structure and design evidence before or alongside implementation.
2. Project Setup and Configuration	7. Creating Project Folder; 8. Server Setup; 9. Backend Structure	Prepare the project workspace, runtime, server, dependencies, environment configuration, and source structure.
3. Backend Development	10. Development and Explanation	Implement and explain authentication, workout operations, search, AI integration, middleware, and error handling.
4. Database Development	11. Configure MongoDB; 12. Create Database Connection; 13. Create Schema and Models	Configure MongoDB Atlas, verify connectivity, and establish User and Workout schemas/models.
5. Project Execution	14. Steps for Project Execution; 15. Output Screenshots; 16. Demo Video	Execute the working system, collect output evidence, and prepare the final demonstration.

6. DEVELOPMENT SEQUENCE

Sequence	Planned Activity	Dependency / Output
1	Confirm project source, documentation, scope, and team responsibilities	Agreed implementation baseline
2	Create project folder and install dependencies	Runnable Node.js project
3	Configure environment variables and database connection	Application configuration ready
4	Start server and verify health endpoint	Basic server execution evidence
5	Configure authentication and verify registration/login/profile	Working protected access
6	Configure User and Workout models and verify MongoDB persistence	Working database layer
7	Implement/verify workout CRUD operations	Core workout management

Sequence	Planned Activity	Dependency / Output
8	Implement/verify workout search	Search by name, category, and date
9	Configure and verify Gemini recommendation and insight operations	Working AI integration
10	Perform API, error, and regression testing	Testing evidence
11	Prepare diagrams, screenshots, documentation, GitHub structure	Submission evidence
12	Record final demonstration and complete Skill Wallet evidence	Final submission package

7. MILESTONES

Milestone	Expected Result
M1 – Planning Baseline	Scope, team responsibilities, requirements, and design baseline confirmed.
M2 – Project Setup	Project folder, Node.js runtime, dependencies, environment configuration, and server startup prepared.
M3 – Database Ready	MongoDB connection and User/Workout models verified.
M4 – Backend Core Ready	Authentication, workout CRUD, search, middleware, and error handling verified.
M5 – AI Integration Ready	Workout recommendation and fitness-insight endpoints configured and tested.
M6 – Testing Complete	Required API tests, error checks, and regression checks documented.
M7 – Documentation Complete	Phase documents, academic documentation, API evidence, diagrams, and repository structure prepared.
M8 – Demonstration Ready	Demo sequence, screen recording, voice explanation, final output, and submission links prepared.

8. TESTING PLAN

Test Area	Planned Checks	Evidence
Server	Start application and verify health response.	Server terminal and health-response evidence.
Authentication	Register, login, and profile access using valid authentication.	API request/response screenshots.
Workout CRUD	Create, retrieve, retrieve by ID, update, and delete workout.	API request/response screenshots.
Workout Search	Search by workout name, category, and date.	API request/response screenshots.
AI Recommendation	Submit age, fitness goal, and experience level.	AI endpoint request/response evidence.
AI Insights	Submit total workouts, average duration, and total calories burned.	AI endpoint request/response evidence.

Test Area	Planned Checks	Evidence
Authorization / Ownership	Verify protected operations require authentication and workout operations remain user-scoped.	Relevant API response evidence.
Error Handling	Check invalid or missing inputs and unavailable/failed service conditions where applicable.	Error response evidence.
Regression	Repeat key authentication, workout, and AI operations after changes.	Final verification evidence.

9. DOCUMENTATION PLAN

Documentation Item	Planned Content
Phase 1 – Brainstorming & Ideation	Project idea, problem identification, objectives, initial scope, concept, and team formation.
Phase 2 – Requirement Analysis	Functional/non-functional requirements, technology stack, system scope, API requirement summary, and limitations.
Phase 3 – Project Design	Architecture, ER diagram, API design, user flow, MVC, DFDs, modules, and security design.
Phase 4 – Project Planning	This planning document, responsibilities, development sequence, milestones, testing, documentation, and demonstration plan.
Phase 5 – Project Development	Actual setup, configuration, source structure, backend development, database integration, AI integration, and selected code evidence.
Phase 6 – Project Testing	Test environment, authentication, workout, AI, error, regression, and results evidence.
Phase 7 – Project Documentation	Academic report, API documentation, README, repository structure, and Skill Wallet evidence mapping.
Phase 8 – Project Demonstration	Demonstration procedure, project purpose, working process, final output, and video link.

10. DEMONSTRATION PLAN

Stage	Planned Demonstration Content
Introduction	State the project name: AI FitTrack – Personalized Fitness Recommendations Powered by AI.
Purpose and Benefits	Explain centralized workout management, authenticated access, workout search, and AI-assisted guidance.
Environment	Show the project structure and start the backend without exposing secrets.
Server Execution	Start the Node.js/Express server and show successful execution.
Authentication	Demonstrate registration, login, and profile access.
Workout Management	Demonstrate create, retrieve, search, update, and delete operations.
AI Functions	Demonstrate workout recommendation and fitness insights with appropriate inputs.
Final Output	Show the generated API responses and explain the completed workflow.
Closing	Summarize the implemented functions and show the final repository/project organization.

11. RISK AND ISSUE MANAGEMENT

Risk / Issue	Planned Response
Environment configuration error	Verify Node.js, npm, environment variables, and server startup before dependent testing.
MongoDB connection failure	Verify Atlas cluster availability, database credentials, connection string, and network access.
Gemini API failure	Verify the configured API key/model and inspect the returned API error before changing application logic.
Authentication failure	Check JWT configuration, login response, token handling, and protected-route behavior.
Incorrect API test data	Use the documented request fields and verify IDs, dates, categories, and numeric values before testing.
Scope expansion	Compare proposed additions against the assigned FitTrack scope before implementing them.
Secret exposure	Keep .env and credentials out of screenshots and GitHub; publish only an environment template.
Documentation mismatch	Cross-check diagrams, tables, screenshots, and claims against the actual implementation before final submission.
Regression after changes	Repeat the affected API tests and the core final test sequence after modifications.

12. GITHUB AND SUBMISSION PLANNING

Repository Area	Planned Content
Root	README.md and project overview.
01_Brainstorming_and_Ideation	Phase 1 document.
02_Requirement_Analysis	Phase 2 document.
03_Project_Design	Phase 3 document and final design diagrams.
04_Project_Planning	This Phase 4 document.
05_Project_Development	Phase 5 document, source-code folder, and development evidence.
06_Project_Testing	Phase 6 document and testing evidence.
07_Project_Documentation	Phase 7 document, academic documentation, API documentation, and Skill Wallet mapping.
08_Project_Demonstration	Phase 8 document and final demo-video link file.

13. PLANNING CONCLUSION

The AI FitTrack project plan provides a controlled sequence from the verified project baseline through setup, database configuration, backend development, AI integration, testing, documentation, and final demonstration. The plan also connects the work to the five Skill Wallet epics and sixteen assigned tasks.